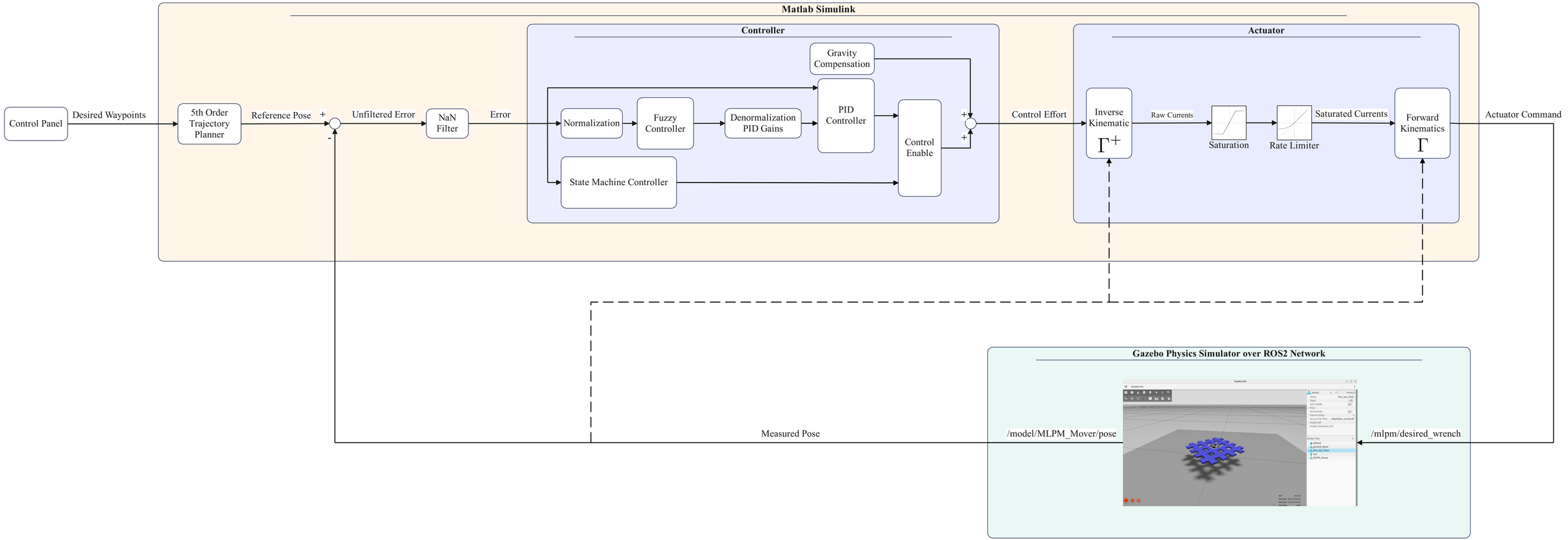

Matlab Simulink
Controller
Actuator
Control Panel
Desired Waypoints
5th Order Trajectory Planner
Reference Pose
+
-
Unfiltered Error
NaN Filter
Error
Normalization
Fuzzy Controller
Denormalization PID Gains
PID Controller
Gravity Compensation
State Machine Controller
Control Enable
+
+
Control Effort
Inverse Kinematic
Γ+
Raw Currents
Saturation
Rate Limiter
Saturated Currents
Forward Kinematics
Γ
Actuator Command
Gazebo Physics Simulator over ROS2 Network
/model/MLPM_Mover/pose
Measured Pose
/mlpm/desired_wrench